WORKED SOLUTION

STEP III 1987, Question 14

Mechanics • Gravitational Collapse

STEP III 1987, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q14

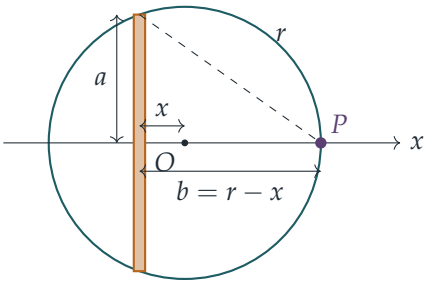
The key idea

A solid sphere is a stack of thin discs, so the force from the whole sphere is an integral of the given disc formula along the axis through the particle. The square root looks unpleasant until we notice that, for a particle on the surface, the expression under it factorises. After that, the cloud question is Newton's second law with the chain rule $\frac{dv}{dt} = v\frac{dv}{dr}$, and the substitution $r = R\cos^2\alpha$ is designed to tidy up the square root that appears when we integrate a second time.

Solution

The force at the surface of a sphere

Put the centre of the sphere at the origin and the particle P at $x = r$ on the x-axis. Slice the sphere into discs perpendicular to the axis. The slice at position x (where $-r \le x \le r$) has thickness δx, radius a with $a^2 = r^2 - x^2$, and lies at distance $b = r - x \ge 0$ from P, on its axis.



The key simplification is in the square root:

$$a^2 + b^2 = (r^2 - x^2) + (r - x)^2 = 2r^2 - 2rx = 2r(r - x),$$

so

$$\frac{b}{(a^2+b^2)^{1/2}} = \frac{r-x}{\sqrt{2r(r-x)}} = \sqrt{\frac{r-x}{2r}}.$$

Each disc pulls P along the axis towards the centre, so the forces all act in the same direction and simply add. Letting $\delta x \to 0$, the total force is

$$\begin{aligned} F &= 2\pi mk\rho \int_{-r}^{r} \left(1 - \sqrt{\frac{r-x}{2r}}\right) \mathrm{d}x \\ &= 2\pi mk\rho \left(2r - \frac{1}{\sqrt{2r}} \int_0^{2r} u^{1/2}\,\mathrm{d}u\right) \qquad (u = r - x) \\ &= 2\pi mk\rho \left(2r - \frac{1}{\sqrt{2r}} \cdot \tfrac{2}{3}(2r)^{3/2}\right) = 2\pi mk\rho \left(2r - \tfrac{4}{3}r\right) = \tfrac{4}{3}\pi mk\rho\, r. \end{aligned}$$

The mass of the sphere is $M = \frac{4}{3}\pi r^3\rho$, so $\frac{4}{3}\pi\rho = M/r^3$ and

$$F = \frac{kmM}{r^2},$$

directed towards the centre, as required.

The equation of motion of the cloud

Model the particles of the cloud as a continuous uniform sphere of mass M, and consider a particle of mass m on its surface at distance r from the centre. By the result above, the only force on it is kmM/r^2 towards the centre. Taking r as the coordinate (positive outwards), Newton's second law gives

$$m\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} = -\frac{kmM}{r^2}.$$

Now $\frac{\mathrm{d}r}{\mathrm{d}t} = -v$, so $\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} = -\frac{\mathrm{d}v}{\mathrm{d}t}$, and by the chain rule

$$\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}v}{\mathrm{d}r}\frac{\mathrm{d}r}{\mathrm{d}t} = -v\frac{\mathrm{d}v}{\mathrm{d}r}.$$

Hence $\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} = v\frac{\mathrm{d}v}{\mathrm{d}r}$, and cancelling m,

$$v\frac{\mathrm{d}v}{\mathrm{d}r} = -\frac{kM}{r^2}.$$

Ask yourself The disc result was for a fixed sphere. Why may we use it for a cloud that is collapsing?

The force at any instant depends only on where the mass is at that instant, so we need the cloud to remain a uniform sphere with the same mass M inside the surface. It does: a particle that starts at rest at radius λR (with $0 < \lambda \leq 1$) has mass $\lambda^3 M$ inside it, and $\lambda r(t)$ satisfies $\ddot{s} = -k\lambda^3 M/s^2$ whenever $r(t)$ satisfies $\ddot{r} = -kM/r^2$. So every particle stays at the same fraction of the surface radius: the cloud shrinks by a uniform scale factor, stays uniform, and no particle overtakes the surface particle. (Mass outside a particle would only matter for interior particles, and none lies outside the surface.)

The collapse from rest

Separate the variables and integrate from $r = R$, where $v = 0$:

$$\tfrac{1}{2}v^2 = \int_R^r -\frac{kM}{s^2}\,\mathrm{d}s = kM\left(\frac{1}{r} - \frac{1}{R}\right) = \frac{kM(R-r)}{rR}.$$

The acceleration is always inwards and the cloud starts at rest, so r decreases and $v \geq 0$ throughout.

Now substitute $r = R\cos^2\alpha$, with α increasing from 0 at $t = 0$ towards $\frac{\pi}{2}$ as r decreases from R towards 0. Then $R - r = R\sin^2\alpha$, so

$$v^2 = \frac{2kM\,R\sin^2\alpha}{R^2\cos^2\alpha} = \frac{2kM}{R}\tan^2\alpha, \qquad v = \left(\frac{2kM}{R}\right)^{1/2}\tan\alpha,$$

taking the positive root since $v \geq 0$ and $\tan\alpha \geq 0$ for $0 \leq \alpha < \frac{\pi}{2}$.

? **Ask yourself** How do we get from v as a function of α to t as a function of α?

Use $v = -\frac{\mathrm{d}r}{\mathrm{d}t}$ and $\frac{\mathrm{d}r}{\mathrm{d}\alpha} = -2R\cos\alpha\sin\alpha$:

$$\frac{\mathrm{d}t}{\mathrm{d}\alpha} = -\frac{1}{v}\frac{\mathrm{d}r}{\mathrm{d}\alpha} = \frac{2R\sin\alpha\cos\alpha}{(2kM/R)^{1/2}\tan\alpha} = \left(\frac{R^3}{2kM}\right)^{1/2} 2\cos^2\alpha = \left(\frac{R^3}{2kM}\right)^{1/2}(1+\cos 2\alpha).$$

Integrating, with $t = 0$ when $\alpha = 0$,

$$t = \left(\frac{R^3}{2kM}\right)^{1/2}\left(\alpha + \tfrac{1}{2}\sin 2\alpha\right),$$

which is the time at which $r = R\cos^2\alpha$, as required.

Checks

- Small times: for small α, $t \approx 2\alpha(R^3/2kM)^{1/2}$ and $R - r \approx R\alpha^2$, so $R - r \approx \frac{1}{2}(kM/R^2)\,t^2$: constant acceleration kM/R^2 from rest, exactly as expected at the start.
- Letting $\alpha \to \frac{\pi}{2}$ gives the total collapse time $\frac{\pi}{2}(R^3/2kM)^{1/2}$, finite even though $v \to \infty$.
- Numerically, summing 200 000 thin discs reproduces kmM/r^2 to a relative error of about 2×10^{-9}. Stepping $\ddot{r} = -kM/r^2$ with $k = 1.7$, $M = 2.3$, $R = 1.9$ to the time given by $\alpha = 0.8$ gives $r = 0.922260$, matching $R\cos^2 0.8 = 0.922260$.

Common slips

- Measuring b from the centre of the sphere instead of from the particle: $b = r - x$, not x.

- Forgetting that v is the *inward* speed, which flips the sign in $\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} = v\frac{\mathrm{d}v}{\mathrm{d}r}$.
- Taking the negative square root for v, which makes the time come out negative.

Beyond STEP

The first part is a special case of Newton's shell theorem: outside a spherically symmetric body, gravity acts as if all the mass were at the centre. University courses prove it neatly with Gauss's law for gravity, $\nabla \cdot \mathbf{g} = -4\pi G\rho$. The collapse time you found is the astrophysicists' *free-fall time*, used to estimate how quickly gas clouds collapse to form stars, and the parametrisation by α is the same one that describes radial Kepler orbits. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats central forces and the shell theorem in this spirit.

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